



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Mid-term Exam: : Trimester: Fall 2025

Course Code: CSE 1111, Course Title: STRUCTURED PROGRAMMING LANGUAGE

Time: 1 hour 30 min Total Marks: 30

(Unfair means in the exam will be taken seriously as per UIU disciplinary rules)

Answer all the Questions. Numerical values in the right margin indicate full marks.

- 1 a) Show the **manual tracing** of the following code 3

```
#include<stdio.h>
int main() {
int sum=23, i=19;
    i=i/4;
    sum=sum%i;
    printf("sum=%5d i=%d\n", sum, i--);
    sum+=i++;
    printf("sum=%5d i=%d", --sum, ++i);
    return 0;
}
```
- b) Find **output** of the following code segment 3

```
int num=45, sum=25;
printf("%d %d\n", num, sum);
switch((num+sum)%3+1){
    case 1:
    case 2: sum=sum+num;
            printf("%d %d\n", num-2, sum);
    case 3:
            printf("%d\n", sum+5); break;
    default:
            printf("%d\n", num+10);
}
printf("%d %d", num, sum);
```
- 2 a) **Rewrite** the code segment using “switch” without changing the logical meaning 4

```
int a, b, c, result=3;
scanf("%d%d%d", &a, &b, &c);
if (result > 0 && result <3) {
    printf("a = %d\n", a);
}
else if (result >= 5) {
    printf("b = %d\n", b);
}
else
    printf("c = %d\n", c);
```
- b) **Manually trace** the following code segment 4

```
int R=0, a=964, i=100;
printf("%d %d\n", R, i);
while(a>0){
    R=a/i;
    printf("%d %d\n", R, i);
    a=a%i;
    i=i/10;
}
```

- 3 a) Draw a **flow chart** for the given code segment

4

```
int num=2,count=0;
do{
    printf("%d\n", num);
    num=num+2;
    if(num%2==0){
        printf("%d\n", num++);
    }
    ++count;
}while(count<=5);
```

- b) Write a **complete program** that prints the difference of two consecutive digits of an input integer number from the right most digit.
For example, num=8396

Difference	Output
9-6=3	3
3-9=-6	6
8-3=5	5

- 4 a) Write a **complete program** that stores input integer data into an array of size n, and makes all the numbers in the array even if it is odd and odd if it is even. Finally, the program prints all the elements in the array.

For example, n=5

Input Array A: 10 33 67 56 98

New Array A: 9 34 68 55 97 (After modification)

Output data: 9 34 68 55 97

- b) **Manually trace** the following code segment

4

```
int i, j, k=11, p=1;
int A[3][4]={12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1};
for(i=0; i<3; i++){
    for(j=0; j<4; j++){
        if(k>=1){
            A[i][j]=A[i][j]-k;
            k=k-2;
        }
        else{
            A[i][j]=A[i][j]+p;
            p=p+2;
        }
        printf("%d ", A[i][j]);
    }
    printf("\n");
}
```